

Research Notes on Random forest

Random forest

>> Section 1

or by taking the plurality vote in the case of classification trees. This bootstrapping procedure leads to better model performance because it decreases the variance of the model, without increasing the bias. This means that while the predictions of a single tree are highly sensitive to noise in its training set, the average of many trees is not, as long as the trees are not correlated. Simply training many trees on a single training set would give strongly correlated trees (or even the same tree many times, if the training algorithm is deterministic); bootstrap sampling is a way of de-correlating the trees by showing them different training sets. Additionally, an estimate of the uncertainty of the prediction can be made as the standard deviation of the predictions from all the individual regression trees on x' :

>> Section 2

Consistency results Assume that $Y = m(\mathbf{X}) + \varepsilon$, where ε is a centered Gaussian noise, independent of $\mathbf{X}$, with finite variance $\sigma^2 < \infty$. Moreover, $\mathbf{X}$ is uniformly distributed on $[0, 1]^d$ and m is Lipschitz. Scornet proved upper bounds on the rates of consistency for centered KeRF and uniform KeRF.

>> Section 3

Consistency of centered KeRF Providing $k \rightarrow \infty$ and $n/2k \rightarrow \infty$, there exists a constant $C_1 > 0$ such that, for all n ,

>> Section 4

$\mathbb{P} \left[a_n \leq N_n(\mathbf{x}, \Theta) \leq b_n \mid \mathcal{D}_n \right] \geq 1 - \varepsilon_n/2$, where $a_n \leq b_n$ and $\varepsilon_n \rightarrow 0$ as $n \rightarrow \infty$.

>> Section 5

$K_{cc}(\mathbf{x}, \mathbf{z}) = \sum_{j=1}^d K_j(\mathbf{x}, \mathbf{z})$, for all $\mathbf{x}, \mathbf{z} \in [0, 1]^d$.

>> Section 6

Uniform KeRF Uniform KeRF is built in the same way as uniform forest, except that predictions are made by $m \sim M$, $n(\mathbf{x}, \Theta_1, \dots, \Theta_M)$, the corresponding kernel function, or connection function is

>> Section 7

Mean decrease in impurity feature importance This approach to feature importance for random forests considers as important the variables which decrease a lot the impurity during splitting. It is described in the book Classification and Regression Trees by Leo Breiman and is the default implementation in sci-kit learn and R. The definition is: unnormalized average importance $(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \sum_{j \in T_i} \Delta_i(j)$, where

>> Section 8

In k -NN, $W(\mathbf{x}_i, \mathbf{x}') = \frac{1}{k}$ if $\mathbf{x}_i$ is one of the k points closest to $\mathbf{x}'$, and zero otherwise. In a tree, $W(\mathbf{x}_i, \mathbf{x}') = \frac{1}{k'}$ if $\mathbf{x}_i$ is one of the k' points in the same leaf as $\mathbf{x}'$, and zero otherwise. Since a forest averages the predictions of a set of m trees with individual weight functions W_j , its predictions are $\hat{y} = \frac{1}{m} \sum_{j=1}^m W_j(\mathbf{x}_i, \mathbf{x}') y_i$.

>> Section 9

$m \sim M$, $n(\mathbf{x}, \Theta_1, \dots, \Theta_M) = \frac{1}{M} \sum_{j=1}^M N_n(\mathbf{x}, \Theta_j)$, where $N_n(\mathbf{x}, \Theta_j)$ is the number of points in the training set that are classified as $\mathbf{x}$ by the tree Θ_j .

>> Section 10

Prefiltering: Eliminate features that are mostly just noise. Enriched Random Forest (ERF): Use weighted random sampling instead of simple random sampling at each node of each tree, giving greater weight to features that appear to be more informative. Tree-weighted random forest (TWRF): Give more weight to more accurate trees.

>> Section 11

$|m(\mathbf{x}) - m(\mathbf{x})| \leq b_n - a_n$, where a_n and b_n are the lower and upper bounds of the prediction.

Research Notes on Random forest

Random forest

$$\|\mathbf{m}_{M,n}(\mathbf{x}) - \mathbf{f}(\mathbf{x})\| \leq \frac{\|\mathbf{b}_n - \mathbf{a}_n\|}{\|\mathbf{a}_n\|} \|\mathbf{m}_{M,n}(\mathbf{x})\|$$

>> Section 12

Variants Instead of decision trees, linear models have been proposed and evaluated as base estimators in random forests, in particular multinomial logistic regression and naive Bayes classifiers. In cases that the relationship between the predictors and the target variable is linear, the base learners may have an equally high accuracy as the ensemble learner.

>> Section 13

$$\sigma = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (f_b(\mathbf{x}') - \bar{f})^2}$$

>> Section 14

Relationship to nearest neighbors A relationship between random forests and the k-nearest neighbor algorithm (k-NN) was pointed out by Lin and Jeon in 2002. Both can be viewed as so-called weighted neighborhoods schemes. These are models built from a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ that make predictions $\hat{y}$ for new points $\mathbf{x}'$ by looking at the "neighborhood" of the point, formalized by a weight function W : $\hat{y} = \sum_{i=1}^n W(\mathbf{x}_i, \mathbf{x}') y_i$. Here, $W(\mathbf{x}_i, \mathbf{x}')$ is the non-negative weight of the i 'th training point relative to the new point $\mathbf{x}'$ in the same tree. For any $\mathbf{x}'$, the weights for points $\mathbf{x}_i$ must sum to 1. Weight functions are as follows: